A Battery Equalization Technique Based on Ćuk Converter Balancing for Lithium Ion Batteries

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***Abstract*—In this paper, an active cell equalization technique based on Ćuk converter balancing for lithium ion (Li-ion) batteries is investigated. The circuit equalizes eight cells, which are connected in series in a battery pack. In Electrical Vehicles (EV), batteries act an important role, due to the fact that the most expense in EV is the cost of the batteries. In a battery pack, voltage differences always exist due to charging and discharging process, Therefore, a balancing circuit required to ensure that all cells are in the same voltage level. There are many balancing methods, among them Ćuk converter balancing has fast balancing time. The conventional Ćuk type uses (2n-1) switches while this proposed circuit requires (n) switches, thus the number of switches is reduced and provides less losses. The simulation results are accomplished to perform the viability of the system.**

***Index Terms*—Balancing circuit, EV, Li-ion, Ćuk converter.**

I. Introduction

In recent years, the demand for electrical vehicles is growing fast, due to increasing air pollution, global warming and limitation of fossil fuels in the world. Thus, electric vehicles act an important role in transportation in the future because they can achieve low pollution and low noise. The cost of electrical vehicles depends on many aspects but the most expense in it is the cost of the batteries. Nowadays, lithium ion based batteries are the most common electrical energy storage device in vehicles. However, the capacity of battery is gradually reduced due to erosion, passivation, out gassing, temperature, decomposition of materials [1]. They are widely used in many application such as electrical vehicles, and motor bikes [2], [3], because of their advantages such as high density, low self-discharge rate, lower weight, higher safety, no memory effect [4], [5]. In industrial application, lithium ion batteries are connected in series, because the open circuit voltage of the battery is low [6]. However, voltage differences exist in a battery pack, along the charging and discharging cycles. These differences are due to electrical and chemical characteristic, unsymmetrical demotion with aging, internal impedance and unequal temperature distribution [7], which reduces the life cycle of the battery [8]. The voltage imbalance is a main aspect to degenerate the proficiency and validity of the battery pack due to the reduction in the usable capacity and the explosion hazard because of overcharging. Therefore, battery management system is required to increase the life cycle and available capacity of the battery. It detects battery voltage, battery charge, battery temperature, discharge current and has over voltage protection and some other tasks. A large number of battery balancing methods have been proposed in the literature [9]–[11]. They are divided into two categories such as, passive equalizing techniques and active equalizing techniques. The passive balancing method connects resistors in parallel to each cell and dissipate energy as heat by using resistors [12]. It is only suitable for lead acid batteries because they can be over charged. The advantages are, low cost, easy implementation, however, the disadvantage is a low efficiency. On other hand, active balancing methods, equalize voltage of the battery cells by transferring charge from the higher cells to the lower cells. They are divides in a few categories such as, switches capacitor, inductor, transformer, buck-boost, and Ćuk converter balancing. The capacitor shuttling balancing method is an active balancing technique that uses capacitors as external energy storage for transferring the energy between the cells [13]. This method has long equalization time compared to other active methods. However, resonant switched capacitor balancing improves the balancing time [9]. The inductor balancing method transfers energy from a cell to other cell by using inductors. It has fast equalization time and low cost as compared with capacitor balancing [10], [14], however, the main disadvantages of the conventional method is long balancing time as compared to transformer balancing, because the energy transfer from the first cell to the last cell is long [15]. The transformer balancing is an active method to transfer energy from one cell to another cell. It has fast equalization time but because of non-uniform turn-ratio of secondary winding leakage inductance, the voltages of the secondary windings are not equal [16], [17]. The Ćuk converter balancing circuit has high efficiency with non dissipative currents and a bidirectional energy transferring capability [18].

II. The proposed Ćuk converter balancing circuit

In this section, the proposed Ćuk converter balancing circuit for lithium-ion batteries will be investigated. As compared with the conventional Ćuk balancing method, which requires (2n-1) switches for n cells, the proposed method uses (n)

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switches and n+1 inductors for n number of cells, that results in less components and losses as shown in Fig. 1 and 2. This circuit equalizes eight cells in a battery string connected in series. It has a buck-boost characteristic using a simple pulse with modulation (PWM) of 50% duty cycle. The operational principle has two main stages, in the first stage, switches Q1, Q3, Q5 and Q7 will be turned on, while the other switches are off. In the next stage, switches Q2, Q4, Q6 and Q8 will be turned on while the others will be turned off.

In order to simplify the operational principle, a circuit with two cells is investigated as shown in Fig. 2. In first stage, when Q1 will be turned on while Q2 is turn off, the equivalent equation can be written as

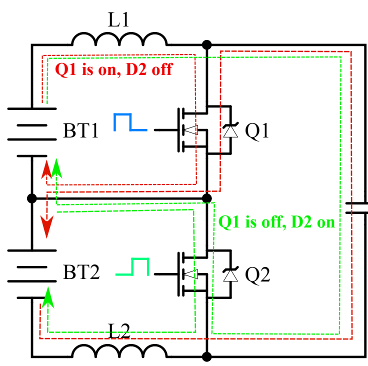

Fig. 1. Simplified Ćuk converter circuit with two cells

$$V_{\text{cell1}} = L_1\frac{di_{L1}}{dt} \tag{1}$$

$$V_{\text{cell2}} = -L_2\frac{di_{L2}}{dt} + V_{C1} \tag{2}$$

In next mode, when Q1 is turned off and the body diode of Q2 is turned on, the equivalent equation can be expressed as

$$V_{\text{cell1}} = L_1\frac{di_{L1}}{dt} + V_{C1} \tag{3}$$

$$V_{\text{cell2}} = -L_2\frac{di_{L2}}{dt} \tag{4}$$

Applying Volt-sec balance across L1

$$V_{\text{cell1}}DT + (V_{\text{cell1}} - V_{C1})(1-D)T = 0 \tag{5}$$

Similarly, Volt-sec balance across L2

$$(-V_{\text{cell2}} + V_{C1})DT + (-V_{\text{cell2}})(1-D)T = 0 \tag{6}$$

Expression for average inductor current can be obtained from charge balance of C1 as

$$i_{L1}(1-D)T_s - i_{L2}DT_s = 0 \tag{7}$$

Where D is duty cycle and T_s is switching frequency. The average inductor current for L_1 and L_2 can be expressed as

$$i_{L1} = [\frac{1}{2}(\frac{V_{\text{cell1}}}{L_1}D^2 + \frac{-V_{C1}+V_{\text{cell1}}}{L_1}(1-D)^2)]T_s \tag{8}$$

$$i_{L2} = [\frac{1}{2}(\frac{V_{C1}-V_{\text{cell2}}}{L_2}D^2 + \frac{-V_{\text{cell2}}}{L_2}(1-D)^2)]T_s \tag{9}$$

Similarly, for the next stage, when Q1 is turned off and Q2 is turned on, the equivalent equation can be expressed as before.

III. Simulation Results

In this section, the simulation results with MATLAB are expressed in order to testify the proposed topology. All the switches are N-channel MOSFETs with body diodes and they are triggered by Pulse Width Modulation (pwm), created by pulse generator with a synchronous pattern and the duty ratio of 50%, as shown in Fig.3.

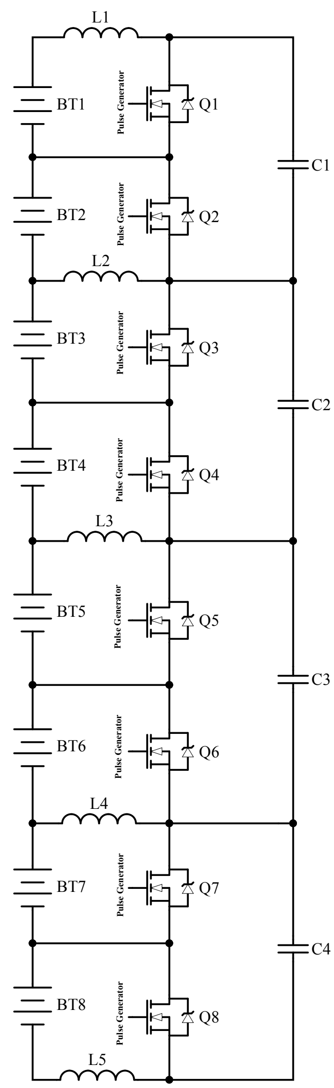

Fig. 2. Schematic of the proposed Ćuk converter balancing circuit

The gate to source voltage (Vgs) of the odd switches is a blue, while the gate to source voltage for even switches is a

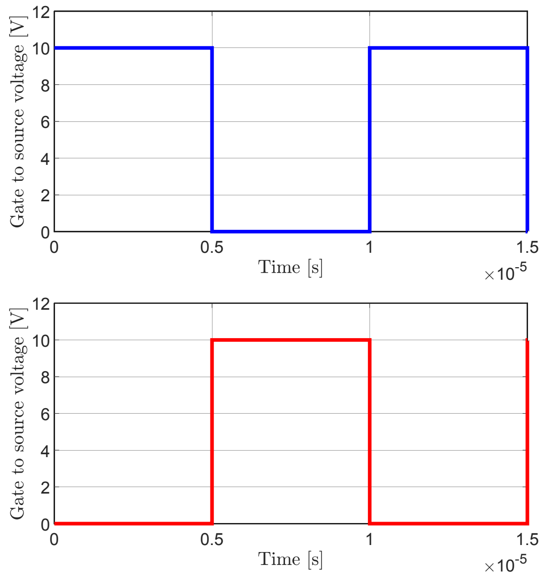

Fig. 3. Gate to source voltage (Vgs) of the MOSSFETs

red. The switching frequency is set to 100kHz, the capacitor values are 100μF and the inductor values are 10μH. The battery voltages are chosen from 3.1 to 3.6 volts to present significant voltage difference of the cells. The batteries are modeled with capacitors with the value of 0.5F. The cell voltages of the proposed circuit is shown in Fig. 4. As it can be seen in the figure, the balancing time is fast with the significant voltage differences. At the time 0.2 s all cells are equalized.

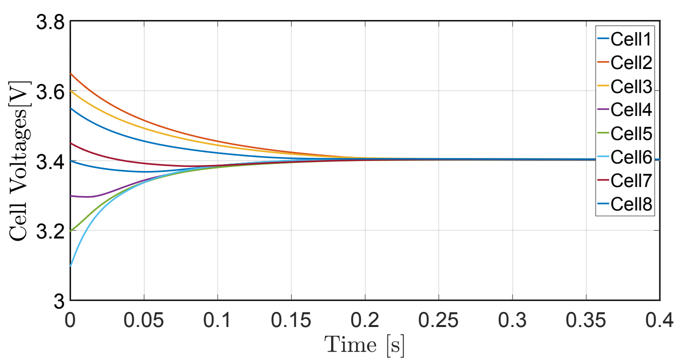

Fig. 4. Cell voltages of the proposed circuit (3.1V to 3.6 V imbalances)

In Fig.5, the inductor voltage waveforms L_1-L_5 in steady state are shown. It has been known that, the average inductor voltage in a period is zero, which can be observed in the figure. The inductor current waveforms are presented in Fig. 6. The figure indicates that, the inductors with negative current, belong to the cells with charging process and the positive currents belong to the cells which are discharging. In order to see the inductor current waveforms in steady state, Fig.7 is presented, this figure shows that all currents are in the same value, it means that, all cells are equalized. If the inductor currents are not at the same level, that means some cells have higher voltages while the others have lower voltages, which

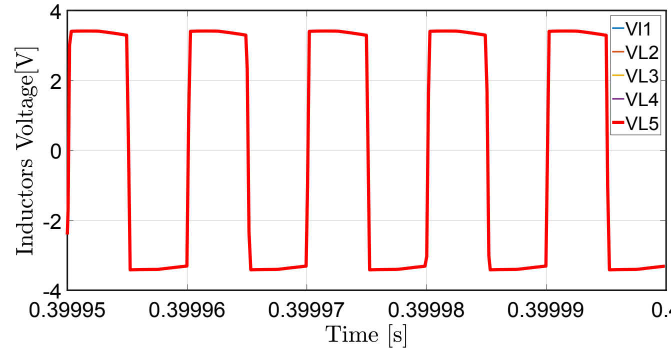

Fig. 5. Inductor voltage waveforms(L_1-L_5) in steady state

means the battery cells are not equalized. At the end, capacitor

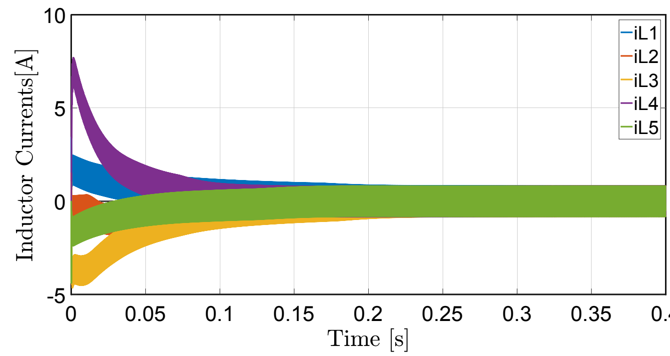

Fig. 6. Inductor current waveforms (L_1-L_5)

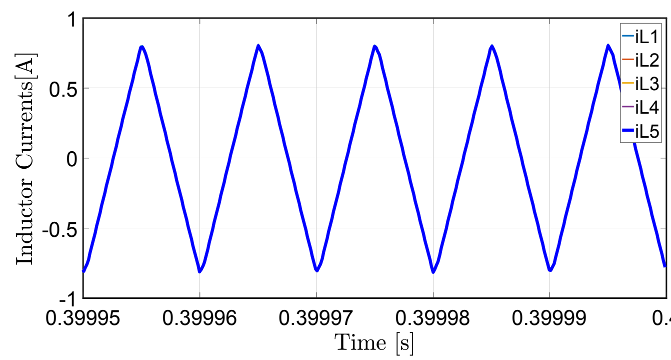

Fig. 7. Inductor current waveforms (L_1-L_5) in steady state

voltage waveforms and capacitor current waveforms (in steady state) are shown in Fig. 8 and Fig. 9 respectively. It can be observed that, the average capacitor voltage values are twice than a voltage of one cell. This can be explicated by having a KVL around larger loop (through cell1, L1, C, L2 and cell2) in Fig. 2, Therefore, the KVL equation can be written as

$$-V_{\text{cell1}} + \bar{V}_{L1} + \bar{V}_C - \bar{V}_{L2} - V_{\text{cell2}} = 0 \qquad (10)$$

However, as it known, the average inductor voltage is zero, therefore $\bar{V}_{L1}$ and $\bar{V}_{L2}$ will be zero, thus according to Eg. 10, the average capacitor voltage $\bar{V}_C$ can be rewritten as

$$\bar{V}_C = V_{\text{cell1}} + V_{\text{cell2}} \qquad (11)$$

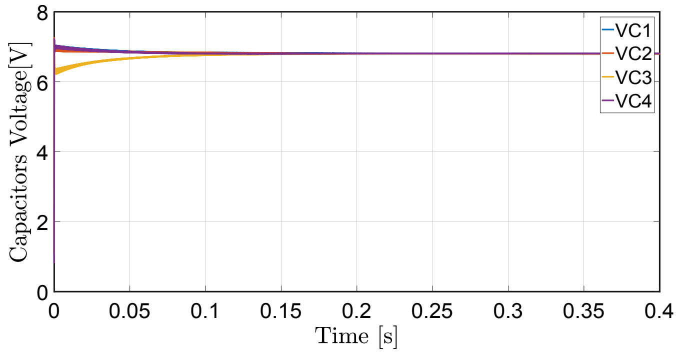

Fig. 8. Capacitor voltage waveforms C_1-C_4

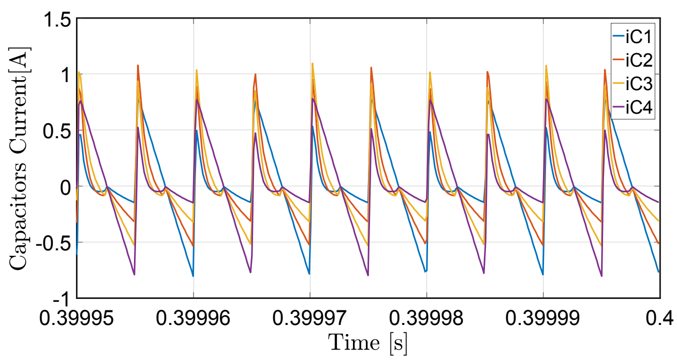

Fig. 9. Capacitor current waveforms C_1-C_4 in steady state

IV. Conclusion

A special Ćuk converter balancing circuit for lithium based batteries in electrical vehicles is investigated in this paper. In electrical vehicles, batteries play an important role due to the fact that, the cost of an electrical vehicle depends on the batteries, thus the batteries with longer life cycle are desired. Therefore, a battery management system is required to increase the life cycle of the batteries. In batteries the cell voltages will differ due to charging and discharging process, thus a balancing circuit is needed to ensure that all cells are at the same voltage level. Among existing cell balancing methods Ćuk converter balancing has a feature of fast balancing time and less costs. In this paper, eight cells in series in a battery pack is considered. This method uses only n switches per n cells. All switches are N-channel MOSFETs (low $R_{DS(on)}$), with body diodes and they are triggered with PWM in synchronous patterns with 50% duty cycle. Therefore, the number of components is reduced and results in less losses and more efficient. The proposed circuit is tested with the significant cell voltage differences to have voltage imbalances in order to verify the proposed topology. The simulation results with MATLAB software are performed to testify the variability of the proposed circuit.

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